448 CARDS

Flashcard deck

FOUNDATIONS AND HOW TO READ A CALL · 1

SIP

Application signaling for sessions. The base specification is RFC 3261.

RFC 3261

FOUNDATIONS AND HOW TO READ A CALL · 2

SDP

Describes media. Offer/answer is RFC 3264.

RFC 3264

FOUNDATIONS AND HOW TO READ A CALL · 3

RTP

Carries the media packets after the answer names the port.

RFC 3550

FOUNDATIONS AND HOW TO READ A CALL · 4

100 Trying

Hop-by-hop provisional. A stateful proxy generates it and does not relay one.

RFC 3261 §16

FOUNDATIONS AND HOW TO READ A CALL · 5

5060

Default port for SIP over UDP and SIP over TCP.

RFC 3261 §18

FOUNDATIONS AND HOW TO READ A CALL · 6

5061

Default port for SIP over TLS.

RFC 3261 §18

FOUNDATIONS AND HOW TO READ A CALL · 7

Content-Length

On a stream transport, this is how the parser finds the end of the message.

RFC 3261 §7

FOUNDATIONS AND HOW TO READ A CALL · 8

WS / WSS

WebSocket transport tokens from RFC 7118. The SIP methods do not change.

RFC 7118

FOUNDATIONS AND HOW TO READ A CALL · 9

tel:+15551212

A global telephone URI. The scheme is RFC 3966.

RFC 3966

FOUNDATIONS AND HOW TO READ A CALL · 10

user=phone

The sip user part should be read as a phone number.

RFC 3261

FOUNDATIONS AND HOW TO READ A CALL · 11

E.164

The ITU-T numbering plan. The URI citation in this course is RFC 3966.

RFC 3966

FOUNDATIONS AND HOW TO READ A CALL · 12

Request-URI

Where this hop sends the request. Retargeting changes it and leaves To in place.

RFC 3261 §8

FOUNDATIONS AND HOW TO READ A CALL · 13

Standards Track

The status of RFC 3261. It is the base specification.

RFC 3261

FOUNDATIONS AND HOW TO READ A CALL · 14

Section 20

Header field catalog, including the compact names defined there.

RFC 3261 §20

FOUNDATIONS AND HOW TO READ A CALL · 15

Section 21

Response code catalog.

RFC 3261 §21

FOUNDATIONS AND HOW TO READ A CALL · 16

Section 25

ABNF. Use it when a message looks almost right.

RFC 3261 §25

FOUNDATIONS AND HOW TO READ A CALL · 17

RFC 5411

Informational hitchhiker's guide. A map, not the base spec.

RFC 5411

FOUNDATIONS AND HOW TO READ A CALL · 18

RFC 3262

PRACK and reliable provisional responses.

RFC 3262

FOUNDATIONS AND HOW TO READ A CALL · 19

RFC 4028

Session timers. Not a substitute chapter inside 5411.

RFC 4028

FOUNDATIONS AND HOW TO READ A CALL · 20

RFC 6026

Corrects INVITE 2xx transaction handling after 3261.

RFC 6026

FOUNDATIONS AND HOW TO READ A CALL · 21

IETF Trust

Legal provisions for IETF documents. Read them before copying protocol text.

IETF Trust Legal Provisions

FOUNDATIONS AND HOW TO READ A CALL · 22

Opus

RFC 6716, plus the Opus license page and the IETF IPR disclosures.

RFC 6716

FOUNDATIONS AND HOW TO READ A CALL · 23

G.711

ITU-T recommendation for PCMU and PCMA. Not an IETF RFC.

ITU-T G.711

FOUNDATIONS AND HOW TO READ A CALL · 24

G.729

ITU-T recommendation. Verify patent status. Do not state an expiry date.

ITU-T G.729

FOUNDATIONS AND HOW TO READ A CALL · 25

z9hG4bK

Magic cookie at the start of a branch parameter.

RFC 3261 §8.1.1.7

FOUNDATIONS AND HOW TO READ A CALL · 26

Max-Forwards: 70

The value a UAC SHOULD put on a request it originates.

RFC 3261 §8.1.1.6

FOUNDATIONS AND HOW TO READ A CALL · 27

No To tag

An initial request is outside a dialog, so To has no tag.

RFC 3261 §12

FOUNDATIONS AND HOW TO READ A CALL · 28

RFC 5737

Documentation IPv4 ranges used for the examples in this course.

RFC 5737

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 29

UAC

Creates a request. Defined with the rest of the user-agent rules.

RFC 3261 §8

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 30

Stateful proxy

Transaction state, own Via branch, own 100 Trying.

RFC 3261 §16

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 31

Redirect

3xx and a Contact, then it is out of the call.

RFC 3261

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 32

B2BUA

UAS plus UAC. Expect a second dialog. See RFC 7092.

RFC 7092

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 33

INVITE / ACK / BYE

Set up, complete the handshake, and end a confirmed dialog.

RFC 3261

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 34

CANCEL

Abandon an unanswered INVITE. It is not BYE.

RFC 3261

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 35

401 vs 407

UAS or registrar challenge versus proxy challenge.

RFC 3261 §22

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 36

1xx

Provisional. The transaction is still waiting for a final response.

RFC 3261 §21

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 37

v / f / t / i

Via, From, To, Call-ID.

RFC 3261 §20

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 38

m / l / c / k

Contact, Content-Length, Content-Type, Supported.

RFC 3261 §20

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 39

o and u

Event and Allow-Events. RFC 6665.

RFC 6665

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 40

r and b

Refer-To and Referred-By. RFC 3515.

RFC 3515

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 41

Branch

Transaction identifier. Cookie z9hG4bK plus a unique token.

RFC 3261 §8.1.1.7

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 42

Retransmission

Same message, same branch, sent again by the transaction timer.

RFC 3261 §17

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 43

2xx ACK

New transaction, new branch, end to end. CSeq number matches the INVITE.

RFC 6026

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 44

Timer B

INVITE transaction timeout, 64 times T1. No ACK is generated.

RFC 3261 §17

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 45

Call-ID + tags

The dialog id. Both tags are required once the dialog exists.

RFC 3261 §12

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 46

Early

Provisional response with a To tag. Not confirmed yet.

RFC 3261 §12

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 47

Confirmed

A 2xx to INVITE. BYE is now the teardown.

RFC 3261 §13

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 48

CSeq

Increments for new in-dialog requests. ACK and CANCEL keep the INVITE number.

RFC 3261 §8

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 49

T1 = 500 ms

Default RTT estimate. Timer A starts here on UDP.

RFC 3261 §17

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 50

64*T1

Timer B for INVITE and Timer F for non-INVITE.

RFC 3261 §17

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 51

T2 = 4 s, T4 = 5 s

Retransmit cap, and how long a message may remain in the network.

RFC 3261 §17

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 52

Timer C

Proxy timer for an unanswered INVITE. More than 3 minutes.

RFC 3261 §16

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 53

Via transport

UDP, TCP, TLS, WS, or WSS for this hop.

RFC 3261 §18

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 54

received / rport

Response address and, with RFC 3581, the source port.

RFC 3581

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 55

session protocol sipv2

Dial-peer command that selects SIP for that peer.

Cisco CUBE basic SIP configuration

SIP CORE: MESSAGES, TRANSACTIONS, DIALOGS · 56

ip address trusted list

Sources allowed to present SIP. See the toll-fraud guidance.

Cisco CME toll-fraud prevention

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 57

Non-2xx ACK

Same branch as that hop's INVITE. Generated by the client transaction.

RFC 3261 §17

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 58

2xx ACK

New branch, dialog route set, end to end.

RFC 6026

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 59

CSeq: 1 ACK

Same number as the INVITE. The method token changes to ACK.

RFC 3261 §8

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 60

603 then ACK

Forward the 603. ACK it on this hop. Do not relay the ACK.

RFC 3261 §16

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 61

Record-Route

Inserted by a proxy that must see later requests in the dialog.

RFC 3261 §16

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 62

lr

Required on that URI. It selects loose routing.

RFC 3261 §16

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 63

UAC reverses

Response Record-Route, reversed, becomes the caller's route set.

RFC 3261 §12

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 64

Contact

The remote target. It becomes the mid-dialog Request-URI.

RFC 3261 §12

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 65

Supported: 100rel

The UAC can do reliable provisionals.

RFC 3262

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 66

RSeq

Identifies the reliable provisional.

RFC 3262

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 67

RAck

RSeq, then the CSeq number and method being acknowledged.

RFC 3262

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 68

PRACK

A new transaction. Its success response is 200 OK, not ACK.

RFC 3262

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 69

UPDATE

Session refresh or change. Own transaction. RFC 3311.

RFC 3311

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 70

No ACK

200 OK finishes UPDATE. ACK is reserved for INVITE.

RFC 3311

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 71

Early dialog

UPDATE can run before the first INVITE completes.

RFC 3311

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 72

RFC 6141

Read this before changing the remote target.

RFC 6141

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 73

Session-Expires

Seconds until the session must be refreshed, plus an optional refresher.

RFC 4028

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 74

Min-SE

Minimum interval. Never below 90 seconds.

RFC 4028

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 75

422

Session Interval Too Small. Retry with the Min-SE from the response.

RFC 4028

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 76

refresher

uac or uas. That side sends re-INVITE or UPDATE.

RFC 4028

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 77

CANCEL

Same branch as the unanswered INVITE. CSeq number unchanged.

RFC 3261

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 78

487

Final response to the INVITE that was cancelled. Then ACK.

RFC 3261 §21

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 79

BYE

Ends a confirmed dialog. Higher CSeq. Both tags.

RFC 3261 §12

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 80

re-INVITE

Mid-dialog INVITE. See RFC 6141 if Contact changes the target.

RFC 6141

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 81

Refer-To

The request the recipient should send. Compact form r.

RFC 3515

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 82

202 Accepted

REFER was accepted. Watch NOTIFY for the real result.

RFC 3515

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 83

Replaces

call-id, to-tag, from-tag of the dialog to take over.

RFC 3891

CALL FLOWS THAT OPERATORS ACTUALLY TRACE · 84

sipfrag

A status line in the NOTIFY body, such as SIP/2.0 200 OK.

RFC 3515

REGISTRATION, DNS, AND ROUTING · 85

To on REGISTER

The address-of-record being bound.

RFC 3261 §10

REGISTRATION, DNS, AND ROUTING · 86

Contact

The binding: where requests for that address-of-record can go.

RFC 3261 §10

REGISTRATION, DNS, AND ROUTING · 87

423

Interval Too Brief. Read Min-Expires and retry.

RFC 3261 §21

REGISTRATION, DNS, AND ROUTING · 88

Contact: *

With Expires: 0, removes all bindings for the address-of-record.

RFC 3261 §10

REGISTRATION, DNS, AND ROUTING · 89

NAPTR

Selects the SIP service, such as SIP+D2U or SIP+D2T.

RFC 3263

REGISTRATION, DNS, AND ROUTING · 90

SRV

Host and port for the selected service.

RFC 3263

REGISTRATION, DNS, AND ROUTING · 91

A / AAAA

Addresses for the SRV target.

RFC 3263

REGISTRATION, DNS, AND ROUTING · 92

Numeric IP

No domain lookup. Send to that address.

RFC 3263

REGISTRATION, DNS, AND ROUTING · 93

lr

Loose routing. Pop Route, keep the Request-URI.

RFC 3261 §16

REGISTRATION, DNS, AND ROUTING · 94

Max-Forwards

Starts at 70. Decremented by each proxy.

RFC 3261 §8.1.1.6

REGISTRATION, DNS, AND ROUTING · 95

483

Too Many Hops. The request arrived with Max-Forwards 0.

RFC 3261 §21

REGISTRATION, DNS, AND ROUTING · 96

482

Loop Detected from a Via loop, not from the hop budget.

RFC 3261 §16

REGISTRATION, DNS, AND ROUTING · 97

reg-id

Identifies one flow. A new value is a new flow.

RFC 5626

REGISTRATION, DNS, AND ROUTING · 98

+sip.instance

URN for the UA instance, stable across flows.

RFC 5626

REGISTRATION, DNS, AND ROUTING · 99

CRLF keepalive

Double CRLF on connection-oriented transports. Single CRLF answers.

RFC 5626

REGISTRATION, DNS, AND ROUTING · 100

STUN keepalive

The UDP keepalive in the outbound specification.

RFC 5626

REGISTRATION, DNS, AND ROUTING · 101

Path

Inbound vector stored with the binding. RFC 3327.

RFC 3327

REGISTRATION, DNS, AND ROUTING · 102

Service-Route

Outbound preloaded route from the registrar. RFC 3608.

RFC 3608

REGISTRATION, DNS, AND ROUTING · 103

lr on Path

The edge URI is loose-routed, same as Record-Route.

RFC 3327

REGISTRATION, DNS, AND ROUTING · 104

Preloaded Route

The UA puts Service-Route into Route on a new INVITE.

RFC 3608

REGISTRATION, DNS, AND ROUTING · 105

pub-gruu

Stable URI for one UA instance.

RFC 5627

REGISTRATION, DNS, AND ROUTING · 106

temp-gruu

Instance URI that hides the address-of-record.

RFC 5627

REGISTRATION, DNS, AND ROUTING · 107

+sip.instance

The URN the registrar binds to those GRUUs.

RFC 5627

REGISTRATION, DNS, AND ROUTING · 108

gr

URI parameter that marks a GRUU.

RFC 5627

REGISTRATION, DNS, AND ROUTING · 109

Fork

One incoming INVITE, one client transaction per target.

RFC 3261 §16

REGISTRATION, DNS, AND ROUTING · 110

Early dialog

A To tag on a provisional response. One per branch.

RFC 3261 §12

REGISTRATION, DNS, AND ROUTING · 111

Multiple 200 OK

Forward each. The UAC ACKs each and may BYE the extras.

RFC 3261 §13

REGISTRATION, DNS, AND ROUTING · 112

6xx

Cancel the other pending branches. Do not start new ones.

RFC 3261 §16

SDP AND OFFER/ANSWER · 113

RFC 8866

Current SDP. It obsoletes RFC 4566. Trust the RFC Editor status banner.

RFC 8866

SDP AND OFFER/ANSWER · 114

c=

Connection address where media should be sent.

RFC 8866

SDP AND OFFER/ANSWER · 115

m=

One media stream: type, port, protocol, and payload types.

RFC 8866

SDP AND OFFER/ANSWER · 116

a=

An attribute. After m=, it belongs to that stream.

RFC 8866

SDP AND OFFER/ANSWER · 117

Same count

The answer has exactly as many m-lines as the offer.

RFC 3264

SDP AND OFFER/ANSWER · 118

Same order

Line n in the answer answers line n in the offer.

RFC 3264

SDP AND OFFER/ANSWER · 119

Port 0

Rejects that stream and keeps its position.

RFC 3264

SDP AND OFFER/ANSWER · 120

RFC 6337

Usage examples and the pitfalls of rebuilding SDP.

RFC 6337

SDP AND OFFER/ANSWER · 121

Encoding name

Match opus to opus even when the payload type numbers differ.

RFC 3264

SDP AND OFFER/ANSWER · 122

0 / 8 / 9 / 18

PCMU/8000, PCMA/8000, G722/8000, G729/8000.

RFC 3551

SDP AND OFFER/ANSWER · 123

G722/8000

RTP clock is 8000. Audio sample rate is 16 kHz.

RFC 3551

SDP AND OFFER/ANSWER · 124

fmtp

Parameters for one payload type. Do not guess unknown ones.

RFC 8866

SDP AND OFFER/ANSWER · 125

sendrecv

Send and receive. The usual active call.

RFC 3264

SDP AND OFFER/ANSWER · 126

sendonly

Send, and do not receive. A common hold.

RFC 3264

SDP AND OFFER/ANSWER · 127

recvonly

Receive only. The usual answer to sendonly.

RFC 3264

SDP AND OFFER/ANSWER · 128

inactive

Neither direction. Also a common hold.

RFC 3264

SDP AND OFFER/ANSWER · 129

m=audio 0

Rejects that audio stream. Keep proto and a format.

RFC 3264

SDP AND OFFER/ANSWER · 130

Same index

A rejected line is still answered in its original position.

RFC 3264

SDP AND OFFER/ANSWER · 131

No common codec

Port 0 on that m-line, not a deleted body.

RFC 3264

SDP AND OFFER/ANSWER · 132

Offer port 0

The answer port is also 0.

RFC 3264

SDP AND OFFER/ANSWER · 133

183 + SDP

Usual early-media pattern. Not true of every 183.

RFC 3960

SDP AND OFFER/ANSWER · 134

180

Ringing. Local ringback when this message has no SDP.

RFC 3261 §21

SDP AND OFFER/ANSWER · 135

100rel

Makes a provisional response reliable. PRACK it.

RFC 3262

SDP AND OFFER/ANSWER · 136

Two bodies

Compare 183 SDP with the 200 OK SDP before you lock the port.

RFC 6337

SDP AND OFFER/ANSWER · 137

RFC 6337

Offer/answer usage and the mistakes to stop repeating.

RFC 6337

SDP AND OFFER/ANSWER · 138

sess-version

Increase it when the SDP changes.

RFC 8866

SDP AND OFFER/ANSWER · 139

Delayed offer

No SDP in the INVITE. Offer in the reliable response. Answer in the ACK.

RFC 3264

SDP AND OFFER/ANSWER · 140

Do not rebuild

Edit m-lines in place. Do not drop, swap, or renumber them.

RFC 6337

RTP, CODECS, AND DTMF · 141

V=2

RTP version is 2.

RFC 3550

RTP, CODECS, AND DTMF · 142

Sequence

Plus one for every packet from that source.

RFC 3550

RTP, CODECS, AND DTMF · 143

Timestamp

Media clock. 20 ms of G.711 or G.722 steps by 160.

RFC 3550

RTP, CODECS, AND DTMF · 144

SSRC

32-bit source id chosen by the sender.

RFC 3550

RTP, CODECS, AND DTMF · 145

PT 0

PCMU/8000, static.

RFC 3551

RTP, CODECS, AND DTMF · 146

PT 8

PCMA/8000, static. Not the same as PCMU.

RFC 3551

RTP, CODECS, AND DTMF · 147

PT 9

G722/8000. Sample rate is still 16 kHz.

RFC 3551

RTP, CODECS, AND DTMF · 148

PT 18

G729/8000, static.

RFC 3551

RTP, CODECS, AND DTMF · 149

G.711

PCMU/8000 and PCMA/8000. Read ITU-T G.711. They are not interchangeable.

ITU-T G.711

RTP, CODECS, AND DTMF · 150

G.729

rtpmap G729/8000, static PT 18. Read ITU-T G.729. This course states no licensing result.

ITU-T G.729

RTP, CODECS, AND DTMF · 151

Opus

RFC 6716. Read the RFC and the license.

RFC 6716

RTP, CODECS, AND DTMF · 152

IPR

Check IETF IPR disclosures. This course states no legal result.

IETF IPR disclosures

RTP, CODECS, AND DTMF · 153

SR

Sender report. Used when this participant has sent RTP.

RFC 3550

RTP, CODECS, AND DTMF · 154

RR

Receiver report. Fraction lost, cumulative loss, jitter.

RFC 3550

RTP, CODECS, AND DTMF · 155

Port pair

RTP on an even port, RTCP on the next higher port, by default.

RFC 3550

RTP, CODECS, AND DTMF · 156

Not SIP

200 OK does not carry fraction lost.

RFC 3550

RTP, CODECS, AND DTMF · 157

RFC 4733

telephone-event. Obsoletes RFC 2833. Trust the status banner.

RFC 4733

RTP, CODECS, AND DTMF · 158

Events 0-15

DTMF. Event 16 is flash.

RFC 4733

RTP, CODECS, AND DTMF · 159

Dynamic PT

Match the name telephone-event in SDP.

RFC 4733

RTP, CODECS, AND DTMF · 160

Not INFO

RFC 6086 is a method. application/dtmf-relay is a vendor body.

RFC 6086

RTP, CODECS, AND DTMF · 161

a=ptime:20

20 ms of media per packet. SDP, not RTP.

RFC 8866

RTP, CODECS, AND DTMF · 162

50 and 160

G.711 at 20 ms: 50 packets/s, 160 payload bytes, plus headers.

RFC 3551

RTP, CODECS, AND DTMF · 163

EF

Voice service class in RFC 4594.

RFC 4594

RTP, CODECS, AND DTMF · 164

RFC 3246

Defines the EF per-hop behavior. No router config in this lesson.

RFC 3246

RTP, CODECS, AND DTMF · 165

200 OK

Offer/answer finished. It does not prove both RTP directions.

RFC 3264

RTP, CODECS, AND DTMF · 166

sendonly answer

Answerer sends, does not receive. One-way by negotiation.

RFC 3264

RTP, CODECS, AND DTMF · 167

c= and port

The destination you must capture.

RFC 8866

RTP, CODECS, AND DTMF · 168

SR from one side

The other side is not sending, according to RTCP.

RFC 3550

TLS, SRTP, IDENTITY, AND ABUSE · 169

401

From the UAS. Challenge is WWW-Authenticate. Retry with Authorization.

RFC 3261 §22

TLS, SRTP, IDENTITY, AND ABUSE · 170

407

From the proxy. Challenge is Proxy-Authenticate. Retry with Proxy-Authorization.

RFC 3261 §22

TLS, SRTP, IDENTITY, AND ABUSE · 171

RFC 7616

HTTP digest algorithms. Read the algorithm parameter.

RFC 7616

TLS, SRTP, IDENTITY, AND ABUSE · 172

Stop here

Identify the header. Do not compute a digest in this lesson.

RFC 3261 §22

TLS, SRTP, IDENTITY, AND ABUSE · 173

RFC 3261 §18

SIP over TLS. Same methods. Via transport TLS.

RFC 3261 §18

TLS, SRTP, IDENTITY, AND ABUSE · 174

5061

Usual TLS port in RFC 3261.

RFC 3261 §18

TLS, SRTP, IDENTITY, AND ABUSE · 175

RFC 5922

The certificate must identify the SIP domain.

RFC 5922

TLS, SRTP, IDENTITY, AND ABUSE · 176

Not SRTP

TLS on SIP does not select the media keying method.

RFC 3261 §18

TLS, SRTP, IDENTITY, AND ABUSE · 177

RFC 3711

SRTP. Protects RTP. Profile RTP/SAVP.

RFC 3711

TLS, SRTP, IDENTITY, AND ABUSE · 178

RFC 4568

SDES. a=crypto in SDP. Protect the signaling.

RFC 4568

TLS, SRTP, IDENTITY, AND ABUSE · 179

RFC 5764

DTLS-SRTP. Keys from DTLS. Fingerprint in SDP.

RFC 5764

TLS, SRTP, IDENTITY, AND ABUSE · 180

Per leg

Do not assume two peers share one keying method.

RFC 5764

TLS, SRTP, IDENTITY, AND ABUSE · 181

RFC 3323

Privacy header. id withholds a network-asserted identity.

RFC 3323

TLS, SRTP, IDENTITY, AND ABUSE · 182

PPI

P-Preferred-Identity. A hint from the UA inside the trust domain.

RFC 3325

TLS, SRTP, IDENTITY, AND ABUSE · 183

PAI

P-Asserted-Identity. Trusted networks only. Not a signature.

RFC 3325

TLS, SRTP, IDENTITY, AND ABUSE · 184

Strip

Remove PAI and PPI when the request leaves the trust domain.

RFC 3325

TLS, SRTP, IDENTITY, AND ABUSE · 185

RFC 8224

STIR. Identity header in SIP.

RFC 8224

TLS, SRTP, IDENTITY, AND ABUSE · 186

RFC 8225

PASSporT.

RFC 8225

TLS, SRTP, IDENTITY, AND ABUSE · 187

RFC 8226

Certificates for STIR.

RFC 8226

TLS, SRTP, IDENTITY, AND ABUSE · 188

Levels A/B/C

ATIS SHAKEN. Confirm the current ATIS document. See also the FCC page.

FCC call authentication

TLS, SRTP, IDENTITY, AND ABUSE · 189

Scanning

Unsolicited SIP from addresses you did not provision. Close the exposure.

RFC 3261 §22

TLS, SRTP, IDENTITY, AND ABUSE · 190

Hijack

Unexpected Contact on an address-of-record. Authenticate REGISTER.

RFC 3261 §10

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Toll fraud

Unauthorized calls that run up charges. Authorize the source.

Cisco CME toll-fraud prevention

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FCC page

Call authentication is not a substitute for outbound authorization.

FCC call authentication

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Allow list

Named peers only. See the CME toll-fraud guide for trusted sources.

Cisco CME toll-fraud prevention

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Digest

401 uses Authorization. 407 uses Proxy-Authorization.

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RFC 5922

Certificate name equals the SIP domain.

RFC 5922

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Per-leg SRTP

Do not copy keying from one peer to another.

RFC 3711

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Embedded

Via, Contact, and SDP c= carry addresses NAT does not rewrite.

RFC 3261

NAT, ICE, AND THE ALG PROBLEM · 198

c=

Media destination. A private value is non-routable on the public Internet.

RFC 8866

NAT, ICE, AND THE ALG PROBLEM · 199

200 OK

Signaling can succeed while RTP still cannot go where c= points.

RFC 3264

NAT, ICE, AND THE ALG PROBLEM · 200

RFC 5737

203.0.113.10 and the other lab addresses are documentation ranges.

RFC 5737

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RFC 3581

rport. Symmetric SIP response routing.

RFC 3581

NAT, ICE, AND THE ALG PROBLEM · 202

received

Source address the server saw.

RFC 3581

NAT, ICE, AND THE ALG PROBLEM · 203

rport value

Source port the server saw.

RFC 3581

NAT, ICE, AND THE ALG PROBLEM · 204

Not SDP

rport does not rewrite c=.

RFC 3581

NAT, ICE, AND THE ALG PROBLEM · 205

RFC 8489

Current STUN. Obsoletes RFC 5389. Trust the status banner.

RFC 8489

NAT, ICE, AND THE ALG PROBLEM · 206

Binding

Request that asks what address the server sees.

RFC 8489

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XOR-MAPPED-ADDRESS

The reflexive address in the success response.

RFC 8489

NAT, ICE, AND THE ALG PROBLEM · 208

Not SIP

A STUN server does not store a SIP Contact.

RFC 8489

NAT, ICE, AND THE ALG PROBLEM · 209

RFC 8656

Current TURN. Obsoletes RFC 5766. Trust the status banner.

RFC 8656

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Allocate

Reserves a relayed transport address.

RFC 8656

NAT, ICE, AND THE ALG PROBLEM · 211

Relay

Peers send media to the relay. TURN forwards it.

RFC 8656

NAT, ICE, AND THE ALG PROBLEM · 212

Not SIP

Allocate is not REGISTER and not an SDP rewrite.

RFC 8656

NAT, ICE, AND THE ALG PROBLEM · 213

RFC 8445

Current ICE. Obsoletes RFC 5245. Trust the status banner.

RFC 8445

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host

Local interface.

RFC 8445

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srflx / relay

STUN reflexive address, and TURN relay.

RFC 8489

NAT, ICE, AND THE ALG PROBLEM · 216

prflx

Discovered during a connectivity check.

RFC 8445

NAT, ICE, AND THE ALG PROBLEM · 217

ALG

In-path SIP and SDP rewrite. Often corrupts the call.

RFC 8866

NAT, ICE, AND THE ALG PROBLEM · 218

Length

If Content-Length does not match the body, the edit is corrupt.

RFC 3261

NAT, ICE, AND THE ALG PROBLEM · 219

TLS

The ALG cannot rewrite what it cannot see.

RFC 3261 §18

NAT, ICE, AND THE ALG PROBLEM · 220

Disable it

Turn the ALG off. Do not stack a second hidden editor.

RFC 3261

NAT, ICE, AND THE ALG PROBLEM · 221

Latch

Learn the remote RTP address from the first packet instead of trusting c=.

RFC 5853

NAT, ICE, AND THE ALG PROBLEM · 222

RFC 5853

SBC requirements. Not an IETF algorithm for latching.

RFC 5853

NAT, ICE, AND THE ALG PROBLEM · 223

Authenticated

Latch only after signaling is accepted, and only toward the expected side.

RFC 3261 §22

NAT, ICE, AND THE ALG PROBLEM · 224

ICE

RFC 8445 is the specified candidate check when both ends use it.

RFC 8445

SESSION BORDER CONTROLLERS · 225

RFC 5853

Informational requirements from SBC deployments. Not an SBC protocol.

RFC 5853

SESSION BORDER CONTROLLERS · 226

No SBC method

INVITE and OPTIONS stay as RFC 3261 defined them.

RFC 3261

SESSION BORDER CONTROLLERS · 227

Datasheet

Session and transcode limits are platform-specific.

RFC 5853

SESSION BORDER CONTROLLERS · 228

B2BUA

The usual SIP role behind an SBC. Taxonomy is RFC 7092.

RFC 7092

SESSION BORDER CONTROLLERS · 229

B2BUA

UAS plus UAC. Two dialogs for one call.

RFC 7092

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Pure proxy

Forwards the request and adds Via.

RFC 3261 §16

SESSION BORDER CONTROLLERS · 231

Call-ID

Often different on the two legs of an SBC.

RFC 7092

SESSION BORDER CONTROLLERS · 232

RFC 3261

Defines the B2BUA as UAS concatenated with UAC.

RFC 3261

SESSION BORDER CONTROLLERS · 233

Via

Identifies the previous SIP hop. Hide the internal one.

RFC 3261

SESSION BORDER CONTROLLERS · 234

Contact

Dialog target. An SBC usually substitutes its own.

RFC 7092

SESSION BORDER CONTROLLERS · 235

c=

SDP connection address. Anchoring rewrites it.

RFC 3264

SESSION BORDER CONTROLLERS · 236

User-Agent

A common topology leak. Stripping it is a product rule.

RFC 5853

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Anchor

RTP terminates on the SBC. SDP c= names the SBC.

RFC 3550

SESSION BORDER CONTROLLERS · 238

Bypass

SIP through the SBC. Media between endpoints.

RFC 3264

SESSION BORDER CONTROLLERS · 239

Transcode

Decode and encode. Capacity is on the datasheet.

RFC 5853

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telephone-event

RFC 4733 DTMF in RTP.

RFC 4733

SESSION BORDER CONTROLLERS · 241

OPTIONS

Capability and liveness request. RFC 3261.

RFC 3261 §11

SESSION BORDER CONTROLLERS · 242

CSeq method

Must match the start line. OPTIONS stays OPTIONS.

RFC 3261 §8

SESSION BORDER CONTROLLERS · 243

200 OK

The peer answered. The call is a later transaction.

RFC 3261 §21

SESSION BORDER CONTROLLERS · 244

Interval

Configured on the product. Not an RFC constant.

RFC 5853

SESSION BORDER CONTROLLERS · 245

Admission

Refuse new work when the configured budget is exhausted.

RFC 5853

SESSION BORDER CONTROLLERS · 246

503

Service Unavailable. Confirm the product’s actual code.

RFC 3261 §21

SESSION BORDER CONTROLLERS · 247

Datasheet

The only source for this platform’s session limit.

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No CPS formula

Do not invent calls-per-second math.

RFC 5853

SESSION BORDER CONTROLLERS · 249

Active

The unit currently processing the dialogs.

RFC 5853

SESSION BORDER CONTROLLERS · 250

Standby

Ready to take the role. Not a SIP method.

RFC 5853

SESSION BORDER CONTROLLERS · 251

No timer here

Do not invent a failover interval.

RFC 5853

SESSION BORDER CONTROLLERS · 252

Dialog state

A B2BUA holds it. Survival is a product question.

RFC 7092

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CUBE

Cisco Unified Border Element. An IOS XE B2BUA.

Cisco CUBE basic SIP configuration

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IOS XE

The operating system whose voice stack CUBE uses.

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CISCO CUBE, CUCM, AND UNITY CONNECTION · 255

SIP profiles

Header modifications. Guide for 17.6 and later.

Cisco CUBE SIP profiles (IOS XE 17.6+)

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B2BUA

The role RFC 7092 taxonomizes.

RFC 7092

CISCO CUBE, CUCM, AND UNITY CONNECTION · 257

dial-peer voice N voip

Creates the VoIP dial peer numbered N.

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CISCO CUBE, CUCM, AND UNITY CONNECTION · 258

session protocol sipv2

This dial peer speaks SIP.

Cisco CUBE basic SIP configuration

CISCO CUBE, CUCM, AND UNITY CONNECTION · 259

session target ipv4:

The next-hop IPv4 address.

Cisco CUBE basic SIP configuration

CISCO CUBE, CUCM, AND UNITY CONNECTION · 260

192.0.2.40

Carrier address in this synthetic.

RFC 5737

CISCO CUBE, CUCM, AND UNITY CONNECTION · 261

sip-profiles

voice class sip-profiles N holds the rules.

Cisco CUBE SIP profiles (IOS XE 17.6+)

CISCO CUBE, CUCM, AND UNITY CONNECTION · 262

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request INVITE sip-header Diversion remove.

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After 15.4

Inbound profile support, per the use-case technote.

Cisco SIP profile use cases

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CSA tester

Try the profile before loading it.

Cisco Collaboration Solutions Analyzer

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Delayed offer

INVITE without SDP. Offer comes later.

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Early offer

The INVITE already contains the offer.

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early-offer forced

Dial-peer command that forces an outbound offer.

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ACK answer

Used when the offer arrived in the 200 OK.

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SIP trunk

CUCM's adjacency to CUBE or another peer.

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CUCM SIP profile

Not voice class sip-profiles.

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Lua

Normalization scripts exist. Open the admin guide.

Cisco Unified Communications Manager

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MTP required

Confirm the guide for that version before checking it.

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SIP integration

The documented Unity Connection path. Open your version’s guide.

Cisco Unity Connection

CISCO CUBE, CUCM, AND UNITY CONNECTION · 274

Diversion

RFC 5806 redirecting number. Reason unconditional in the lab.

RFC 5806

CISCO CUBE, CUCM, AND UNITY CONNECTION · 275

History-Info

RFC 7044. Use it if the guide asks for it.

RFC 7044

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Not SCCP-by-guess

Do not invent SCCP as the current default.

Cisco Unity Connection

CISCO CUBE, CUCM, AND UNITY CONNECTION · 277

debug ccsip messages

Shows sent and received SIP.

Cisco CUBE basic SIP configuration

CISCO CUBE, CUCM, AND UNITY CONNECTION · 278

sip-profiles debug

debug ccsip feature sip-profiles.

Cisco CUBE SIP profiles (IOS XE 17.6+)

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CSA

SIP Profile Tester in Collaboration Solutions Analyzer.

Cisco Collaboration Solutions Analyzer

CISCO CUBE, CUCM, AND UNITY CONNECTION · 280

Sent / Received

Direction markers around each SIP message.

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realm

Routing and policy context on the Oracle SBC.

Oracle Communications SBC documentation

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session-agent

Named next hop.

Oracle Communications SBC documentation

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in-manipulationid

Which sip-manipulation runs on arrival.

Oracle SBC HMR configuration (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 284

out-manipulationid

Which sip-manipulation runs on the way out.

Oracle SBC HMR configuration (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 285

header-rule

One rule inside a sip-manipulation.

Oracle SBC HMR configuration (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 286

delete

Removes the named header. Lab rule strip-server.

Oracle SBC HMR fundamentals (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 287

store

Keeps a value for a later rule. Run it first.

Oracle SBC HMR fundamentals (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 288

reject

An HMR action. Read the release guide for the status.

Oracle SBC HMR fundamentals (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 289

SWe Edge

Edge product. Family includes SBC 1000 and SBC 2000.

Ribbon SBC SWe Edge

ORACLE, RIBBON, AND AUDIOCODES · 290

SBC Core

A separate family.

Ribbon SBC SWe Edge

ORACLE, RIBBON, AND AUDIOCODES · 291

ACS snapshot

5110, 5210, 5400, 7000, and SWe for that validation.

Ribbon documentation portal

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No invented CLI

Admin guides live on the documentation portal.

Ribbon documentation portal

ORACLE, RIBBON, AND AUDIOCODES · 293

Action Subject

Which header or element the rule touches.

AudioCodes SIP message manipulation (7.6)

ORACLE, RIBBON, AND AUDIOCODES · 294

Action Type

Add, Remove, Modify, prefix, suffix, or Normalize.

AudioCodes manipulation syntax 7.6

ORACLE, RIBBON, AND AUDIOCODES · 295

IP group

Inbound or outbound assignment of the set.

AudioCodes SIP message manipulation (7.6)

ORACLE, RIBBON, AND AUDIOCODES · 296

Release menu

The clicks change. Use the 7.6 manual you are running.

AudioCodes SIP message manipulation (7.6)

ORACLE, RIBBON, AND AUDIOCODES · 297

CUBE line

request INVITE sip-header NAME remove.

Cisco CUBE SIP profiles (IOS XE 17.6+)

ORACLE, RIBBON, AND AUDIOCODES · 298

Oracle block

header-rule, header-name, action delete.

Oracle SBC HMR fundamentals (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 299

AudioCodes row

Action Type Remove on an IP group.

AudioCodes SIP message manipulation (7.6)

ORACLE, RIBBON, AND AUDIOCODES · 300

Do not paste across

Each grammar is parsed by its own product.

Oracle SBC HMR configuration (10.1)

ORACLE, RIBBON, AND AUDIOCODES · 301

PAI

P-Asserted-Identity, RFC 3325.

RFC 3325

ORACLE, RIBBON, AND AUDIOCODES · 302

Privacy

RFC 3323. id asks the network to withhold identity.

RFC 3323

ORACLE, RIBBON, AND AUDIOCODES · 303

Diversion

RFC 5806 redirecting header.

RFC 5806

ORACLE, RIBBON, AND AUDIOCODES · 304

History-Info

RFC 7044. Map it only as the interop guide says.

RFC 7044

ORACLE, RIBBON, AND AUDIOCODES · 305

RFC 5922

SIP domain certificates.

RFC 5922

ORACLE, RIBBON, AND AUDIOCODES · 306

Teams Contact

FQDN. An IP draws 403.

Direct Routing SIP protocol

ORACLE, RIBBON, AND AUDIOCODES · 307

Teams wildcard

CN or SAN wildcards are supported for Direct Routing.

Plan Direct Routing

ORACLE, RIBBON, AND AUDIOCODES · 308

Webex LGW wildcard

Not supported for Local Gateway.

Get started with Local Gateway

CLOUD VOICE AND CCAAS SIP · 309

pstnhub

sip, sip2, and sip3.pstnhub.microsoft.com.

Direct Routing SIP protocol

CLOUD VOICE AND CCAAS SIP · 310

5061

Direct Routing signaling port.

Direct Routing SIP protocol

CLOUD VOICE AND CCAAS SIP · 311

Verified domain

Not a name ending in onmicrosoft.com.

Plan Direct Routing

CLOUD VOICE AND CCAAS SIP · 312

No RFC 5923

The SIP proxy does not support connection aliasing.

Direct Routing definitions and RFCs

CLOUD VOICE AND CCAAS SIP · 313

OPTIONS 200

Marks the Direct Routing SBC healthy.

Direct Routing OPTIONS and TLS

CLOUD VOICE AND CCAAS SIP · 314

CSeq

Method token matches the request. OPTIONS, not INVITE.

RFC 3261 §8

CLOUD VOICE AND CCAAS SIP · 315

Table cell

Checked 2026-10-14: CSeq: 1 INVITE. Do not copy it.

Direct Routing SIP protocol

CLOUD VOICE AND CCAAS SIP · 316

SRTP

Media processors use SRTP. RFC 3711.

RFC 3711

CLOUD VOICE AND CCAAS SIP · 317

SIP TLS

Signaling from an IOS XE Local Gateway to Webex.

Local Gateway on IOS XE

CLOUD VOICE AND CCAAS SIP · 318

SRTP

Media for that IOS XE Local Gateway.

Local Gateway on IOS XE

CLOUD VOICE AND CCAAS SIP · 319

Certificate trunk

How a third-party SBC is supported.

Webex Calling trunks and dial plans

CLOUD VOICE AND CCAAS SIP · 320

ByoPSTN

Bring your own PSTN. Not Cloud Connected PSTN.

Get started with Local Gateway

CLOUD VOICE AND CCAAS SIP · 321

Cisco Calling Plan

One of the three organization-article choices.

Configure Webex Calling PSTN

CLOUD VOICE AND CCAAS SIP · 322

Cloud Connected PSTN

A cloud partner through Webex. Not the Local Gateway.

Configure Webex Calling PSTN

CLOUD VOICE AND CCAAS SIP · 323

Local Gateway

Premises-based PSTN. One connection on the location.

Get started with Local Gateway

CLOUD VOICE AND CCAAS SIP · 324

ByoPSTN

Bring your own PSTN.

Get started with Local Gateway

CLOUD VOICE AND CCAAS SIP · 325

Managed PSTN

Ordinary Connect calls. No customer SIP trunk required.

Amazon Connect telephony

CLOUD VOICE AND CCAAS SIP · 326

SIPREC

External voice into the analytics connector.

Connect conversational analytics connector

CLOUD VOICE AND CCAAS SIP · 327

Voice Connector

Chime SDK SIP trunking service.

Amazon Chime SDK Voice Connector

CLOUD VOICE AND CCAAS SIP · 328

Not a bridge

Carrier through Voice Connector into Connect needs specialist approval.

Voice Connector and Connect

CLOUD VOICE AND CCAAS SIP · 329

BYO SIP

Your ITSP or gateway connects to Five9 SBCs.

Five9 BYO SIP trunking

CLOUD VOICE AND CCAAS SIP · 330

Entitled groups

Self-service exists for entitled BYO SIP trunk groups.

Five9 BYO SIP trunking

CLOUD VOICE AND CCAAS SIP · 331

Termination URI

Must appear in the Request-URI or Twilio drops the request.

Twilio trunk troubleshooting

CLOUD VOICE AND CCAAS SIP · 332

sip not sips

Origination URLs use sip. sips is not supported.

Twilio Elastic SIP Trunking API

CLOUD VOICE AND CCAAS SIP · 333

BYOC Cloud

Public internet SIP. RTP crosses the internet.

Genesys BYOC Cloud overview

CLOUD VOICE AND CCAAS SIP · 334

BYOC Premises

Customer-managed Edge. Hybrid media is allowed.

BYOC Cloud with BYOC Premises

CLOUD VOICE AND CCAAS SIP · 335

December 1, 2026

Genesys Hardware Solution only.

About BYOC Premises

CLOUD VOICE AND CCAAS SIP · 336

CXone ports

5060 and 5061, plus dynamic RTP/SRTP.

NiCE CXone ports

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 337

Call-ID

Same value for the dialog. A B2BUA may use a new one on the other leg.

RFC 3261 §8

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 338

CSeq

Integer plus method. The ACK for an INVITE keeps the INVITE's number.

RFC 3261 §8

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 339

Via branch

Transaction hop id. Look for the z9hG4bK cookie.

RFC 3261 §17

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2xx ACK

End to end. New branch. RFC 6026 corrects the transaction rules around it.

RFC 6026

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 341

180

Ringing. Provisional alerting.

RFC 3261 §21

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 342

183

Session Progress. Provisional, often where early media shows up.

RFC 3261 §21

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 343

487

Request Terminated. The INVITE's answer after CANCEL.

RFC 3261 §21

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 344

606

Not Acceptable. Global. 488 is only this UA.

RFC 3261 §21

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 345

sip

Display filter for SIP packets. Field names live on the SIP reference page.

Wireshark SIP display filter reference

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 346

rtp

Display filter for RTP packets. Field names live on the RTP reference page.

Wireshark RTP display filter reference

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 347

sip.Method

Safe display field for the SIP method.

Wireshark SIP display filter reference

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 348

Confirm CSeq

sip.CSeq.method is common. Confirm it on the SIP display-filter reference.

Wireshark SIP display filter reference

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 349

ccsip messages

IOS voice debug that prints SIP send and receive.

Cisco CUBE basic SIP configuration

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 350

ccapi inout

IOS voice debug of the call-control API. Not a SIP RFC.

Cisco CUBE basic SIP configuration

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 351

Sent / Received

Direction on the debug. The SIP start line comes after it.

RFC 3261

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 352

Synthetic

Lab output in this track is synthetic. Customer logs stay out.

RFC 3261

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 353

488

Not Acceptable Here. This UA. Often codec or SDP.

RFC 3261 §21

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 354

422

Session Interval Too Small. RFC 4028. Read Min-SE.

RFC 4028

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 355

telephone-event

RFC 4733. Offer it on purpose if you need DTMF.

RFC 4733

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 356

Offer/answer

The answer is a subset. No shared audio codec tends to 488.

RFC 3264

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 357

Name mismatch

Handshake fails. No SIP status from the peer.

Direct Routing OPTIONS and TLS

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 358

Untrusted CA

Handshake fails. Fix the trust anchor.

Direct Routing OPTIONS and TLS

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 359

TLS 1.2+

Teams Direct Routing floor in the OPTIONS troubleshooting page.

Direct Routing OPTIONS and TLS

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 360

FQDN Contact

Required toward Teams. An address literal draws 403.

Direct Routing SIP protocol

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Category

Read it on the header. 3261 is Standards Track. 6881 is BCP.

RFC 3261

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Obsoletes

The new RFC names the one it replaces. 8866 and 4566 are the SDP pair here.

RFC 8866

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 363

Updates

A correction that leaves the base in force. Check 6026's header.

RFC 6026

TROUBLESHOOTING LADDERS, CAPTURES, AND DEBUG · 364

rfc-editor.org

The header you cite. Months in this lesson are not invented.

RFC 5411

TOLL FRAUD, E911, RECORDING, AND QOS · 365

Trusted list

Who may present VoIP signaling. CME and SRST each have a page.

Cisco CME toll-fraud prevention

TOLL FRAUD, E911, RECORDING, AND QOS · 366

401 / 407

UAS and proxy digest challenges. Secrets stay unprinted.

RFC 3261 §22

TOLL FRAUD, E911, RECORDING, AND QOS · 367

Numbering plan

Dial-peers match the plan, including whether international destinations are in it.

Cisco CME toll-fraud prevention

TOLL FRAUD, E911, RECORDING, AND QOS · 368

STIR context

FCC call authentication and RFC 8224. A different question from the source list.

FCC call authentication

TOLL FRAUD, E911, RECORDING, AND QOS · 369

trusted list

Addresses allowed to present inbound VoIP. CME page or SRST page.

Cisco CME toll-fraud prevention

TOLL FRAUD, E911, RECORDING, AND QOS · 370

authenticate

The check that uses the list.

Cisco CME toll-fraud prevention

TOLL FRAUD, E911, RECORDING, AND QOS · 371

Two pages

CME and SRST differ. Quote no shared list size.

Cisco SRST trusted IP authentication

TOLL FRAUD, E911, RECORDING, AND QOS · 372

Border pair

198.51.100.30 and 192.0.2.40 in the teaching snippet.

Cisco CME toll-fraud prevention

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urn:service:sos

RFC 5031 service URN for emergency calling.

RFC 5031

TOLL FRAUD, E911, RECORDING, AND QOS · 374

RFC 6881

BCP for emergency calling services.

RFC 6881

TOLL FRAUD, E911, RECORDING, AND QOS · 375

Geolocation

RFC 6442 location conveyance.

RFC 6442

TOLL FRAUD, E911, RECORDING, AND QOS · 376

fcc-mlts

Kari's Law and RAY BAUM's Act. The page wins over a retelling.

FCC MLTS 911 requirements

TOLL FRAUD, E911, RECORDING, AND QOS · 377

7866

Session recording protocol. SRC forks to an SRS.

RFC 7866

TOLL FRAUD, E911, RECORDING, AND QOS · 378

7865

Recording metadata. Read it for fields.

RFC 7865

TOLL FRAUD, E911, RECORDING, AND QOS · 379

Disclosed

Quality monitoring and compliance, with the parties on notice.

RFC 7866

TOLL FRAUD, E911, RECORDING, AND QOS · 380

Retention

A legal question. Outside the RFC.

RFC 7865

TOLL FRAUD, E911, RECORDING, AND QOS · 381

EF

Expedited Forwarding per-hop behavior. RFC 3246.

RFC 3246

TOLL FRAUD, E911, RECORDING, AND QOS · 382

Telephony

The service class RFC 4594 maps to EF.

RFC 4594

TOLL FRAUD, E911, RECORDING, AND QOS · 383

Codepoint

Use the value RFC 3246 recommends. Do not invent a second one.

RFC 3246

TOLL FRAUD, E911, RECORDING, AND QOS · 384

Not a LAN

Queues and campus policy stay in the approved design.

RFC 4594

TOLL FRAUD, E911, RECORDING, AND QOS · 385

OPTIONS

Liveness. Alarm when the trunk stops answering.

RFC 3261 §11

TOLL FRAUD, E911, RECORDING, AND QOS · 386

503

Service Unavailable. Page the server-side class.

RFC 3261 §21

TOLL FRAUD, E911, RECORDING, AND QOS · 387

TLS floor

Toward Teams, 1.2 or newer, or the handshake never yields SIP.

Direct Routing OPTIONS and TLS

TOLL FRAUD, E911, RECORDING, AND QOS · 388

SRS down

Recording-session alarm. RFC 7866. Retention stays outside the RFC.

RFC 7866

TOLL FRAUD, E911, RECORDING, AND QOS · 389

Baseline

The last good OPTIONS or the current snippet, written down.

RFC 3261 §11

TOLL FRAUD, E911, RECORDING, AND QOS · 390

One variable

A single address, codec, or TLS setting.

RFC 3264

TOLL FRAUD, E911, RECORDING, AND QOS · 391

Verify

One signal. A 200, or the test call to +15551212.

RFC 3261

TOLL FRAUD, E911, RECORDING, AND QOS · 392

Rollback

The previous value of that same variable.

Cisco CUBE basic SIP configuration

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 393

Role

What the box does on the ladder: UA, proxy, registrar, B2BUA.

RFC 3261

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B2BUA

UAS plus UAC. RFC 7092. Typical SBC.

RFC 7092

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 395

Document

The cite you can open. A logo is not a document.

RFC 5411

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 396

SDP row

RFC 8866 now. RFC 4566 is the one it moved from.

RFC 8866

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 397

WFM / QM

Operations systems. No SIP RFC in this library.

RFC 7866

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 398

SIPREC

The recording neighbor that is SIP. RFC 7866 and RFC 7865.

RFC 7865

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 399

CRM

Not a SIP element. No invented vendor features.

RFC 3261

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 400

Above the media

Analytics and agents. SIP and RTP still carry the session.

RFC 3550

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 401

Above RTP

Analytics and agents. The packets remain RTP.

RFC 3550

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 402

Dialog

Still Call-ID, CSeq, To tag. RFC 3261.

RFC 3261 §12

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 403

Disclosed fork

SIPREC when the SBC records. RFC 7866.

RFC 7866

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 404

No invented features

Vendor guides or an honest gap. Not a fake RFC.

RFC 3261

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 405

SSCA

SIP School Certified Associate. 20 CEC credits toward BICSI, per the SSCA page.

SSCA Elite

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CLCOR

350-801. Core for CCNP Collaboration and the Collaboration Core specialist.

350-801 CLCOR

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Concentrations

300-815 CLACC, 300-820 CLHCT, or 300-830 CLCCE. One of them, plus the core.

CCNP Collaboration

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MS-721

Collaboration Communications Systems Engineer Associate. Certifications change. Use the link.

MS-721 certification

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Banner

RFC Editor header beats a blog.

RFC 5411

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 410

Cisco page

Exam topics change. The Cisco page wins.

Cisco exams list

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MS-721 link

Certifications change. Use the certification page.

MS-721 certification

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Checked dates

February 3, 2026. April 28, 2026. October 29, 2026. Source check 2026-10-14.

Cisco certification updates

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CUCM

Registrar and trunk call-control point.

Cisco Unified Communications Manager

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 414

UCCX

Call control through CUCM. Not the carrier trunk.

Cisco Unified Contact Center Express

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 415

CUBE or CUCM

Where PSTN SIP lands.

Cisco CUBE basic SIP configuration

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CVP

SIP among CVP, gateways, and CUCM, in that version's guide.

Cisco Unified Customer Voice Portal

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 417

Local Gateway

Premises PSTN pattern for Webex Calling.

Get started with Local Gateway

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 418

Organization PSTN

Configured on the Webex Calling PSTN page.

Configure Webex Calling PSTN

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 419

Placeholder host

example.com stands in so the lesson does not invent the edge FQDN.

Get started with Local Gateway

ECOSYSTEM, CERTIFICATIONS, AND THE UC BRIDGE · 420

Separate product

Webex Contact Center. Check its current design guide.

Configure Webex Calling PSTN

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 421

Call-ID + To tag

Dialog match. No To tag on a brand-new INVITE.

RFC 3261 §12

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 422

Branch

Transaction hop. z9hG4bK.

RFC 3261 §17

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 423

2xx ACK

End to end. RFC 6026.

RFC 6026

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 424

487

INVITE's final response after CANCEL. ACK is hop by hop.

RFC 3261 §21

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 425

488

This UA. Session description. RFC 3261 and RFC 3264.

RFC 3264

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 426

422

RFC 4028. Raise the session interval to Min-SE.

RFC 4028

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 427

telephone-event

RFC 4733. DTMF in the offer you mean.

RFC 4733

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 428

Keying

RFC 3711 plus either RFC 4568 or RFC 5764. Record which.

RFC 4568

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 429

RFC 7092

B2BUA taxonomy. New Call-ID on the other leg is normal.

RFC 7092

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 430

RFC 5853

SBC deployment requirements.

RFC 5853

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 431

Trunk landing

CUBE or CUCM. Not UCCX.

Cisco CUBE basic SIP configuration

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 432

Teams pair

TLS 1.2+ and an FQDN in Contact.

Direct Routing SIP protocol

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 433

Symptom

Silence after connect, Alice at 203.0.113.10. Written first.

RFC 3261

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 434

Ladder

INVITE, 100, 180, 200, end-to-end ACK. One Call-ID.

RFC 6026

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 435

One packet

200 OK SDP. 203.0.113.20, UDP 19642, PCMU.

RFC 3264

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 436

One variable

Capture that port. Config edits wait, and then they are single.

RFC 3550

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 437

Six rows

TLS, codecs, numbering, trusted addresses, OPTIONS, SRTP keying.

RFC 3264

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 438

telephone-event

On the codec row on purpose. RFC 4733.

RFC 4733

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 439

E.164

Number form. tel URI is RFC 3966.

RFC 3966

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 440

Not a dump

Decisions only. No private key, no full dial-peer list.

Cisco CUBE basic SIP configuration

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 441

Path

CUCM, then a certified SBC, then Teams. No invented SKU.

Plan Direct Routing

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 442

CUBE guide

Basic SIP configuration for the CUBE case.

Cisco CUBE basic SIP configuration

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 443

TLS 1.2+

Teams floor. Handshake failures are not SIP codes.

Direct Routing OPTIONS and TLS

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 444

FQDN Contact

sbc.example.com in the lab. An address literal draws 403.

Direct Routing SIP protocol

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 445

SDP

RFC 4566 moved to RFC 8866.

RFC 8866

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 446

ICE

RFC 5245 moved to RFC 8445. Confirm the banner.

RFC 8445

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 447

STUN / TURN

RFC 8489 and RFC 8656 in this library. Predecessors 5389 and 5766.

RFC 8489

INTERVIEWS, INCIDENTS, AND DESIGN DRILLS · 448

Banner

A newer obsoleting RFC that this edition does not cite still wins.

RFC 5411

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